Boolean Logic & Truth Tables

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OCR GCSE Computer Science — **Boolean Logic (J277/02 Section 2.4)**. This topic covers the three fundamental logic gates (AND, OR, NOT), how to construct and complete truth tables for logic expressions, and how to draw and interpret logic circuit diagrams.

Exam questions typically ask you to **complete** truth tables for given expressions, **draw** logic circuits from Boolean expressions, and **identify** the output of a circuit for given inputs. The 2024 exam included completing a truth table for (A AND B) OR C and drawing a circuit for NOT A AND (B OR C).

What you must know

This checklist covers the key exam-visible concepts. You should be able to perform each skill quickly and accurately.

- **State** the function of each logic gate: AND, OR, NOT.
- **Complete** truth tables for expressions using AND, OR, NOT and combinations.
- **Draw** logic circuit diagrams from Boolean expressions using standard gate symbols.
- **Determine** the output of a logic circuit for given input values.
- **Identify** the Boolean expression represented by a given logic circuit.
- **Explain** how brackets affect the order of evaluation in Boolean expressions.

Specification note. OCR J277 Section 2.4 requires understanding of AND, OR, and NOT gates only. You must know the standard symbols for each gate and be able to construct truth tables for expressions combining these gates. NAND, NOR, and XOR gates are **not** required for this specification.

Toolkit

What is Boolean logic?

Boolean logic is a form of algebra where all values are either **TRUE (1)** or **FALSE (0)**. It is named after mathematician George Boole. Computers use Boolean logic to make decisions — every condition in a program (like `if age >= 18`) evaluates to true or false. Logic gates are the physical implementation of Boolean operations in electronic circuits.

The three logic gates

Gate	Symbol shape	Written as	What it does	Real-world analogy
AND	D-shape (flat left, curved right)	A AND B	Output is 1 only if **both** inputs are 1	Two switches in series — both must be ON for current to flow
OR	Curved shield shape (pointed right)	A OR B	Output is 1 if **at least one** input is 1	Two switches in parallel — either can be ON for current to flow
NOT	Triangle with small circle (bubble) at output	NOT A	Output is the **opposite** of the input	An inverter — turns ON to OFF and OFF to ON

Truth table patterns — memorise these

NOT gate (1 input)

A	NOT A
0	1
1	0

Output is always the opposite of input.

AND gate (2 inputs)

A	B	A AND B
0	0	0
0	1	0
1	0	0
1	1	1

Only 1 when **both** inputs are 1.

OR gate (2 inputs)

A	B	A OR B
0	0	0
0	1	1
1	0	1
1	1	1

Only 0 when **both** inputs are 0.

Order of operations (precedence)

When evaluating Boolean expressions without brackets, apply operations in this order:

1. **NOT** — evaluate first (highest precedence)
2. **AND** — evaluate second
3. **OR** — evaluate last (lowest precedence)

Brackets override precedence — always evaluate what's inside brackets first, just like in maths.

Example: `NOT A AND B OR C` is evaluated as `((NOT A) AND B) OR C`

Example: `NOT (A AND B) OR C` — the NOT applies to the **whole** result of (A AND B).

How to complete a truth table

Step 1: List all possible input combinations. For n inputs, there are 2^n rows (2 inputs = 4 rows, 3 inputs = 8 rows).
Step 2: Add columns for any intermediate results (sub-expressions in brackets).
Step 3: Work through each row, evaluating sub-expressions first, then combining for the final output.
Step 4: Double-check by verifying at least one row manually.

Real-world applications of Boolean logic

Boolean logic isn't just abstract theory — it's used everywhere in computing and everyday devices:

Application	Boolean expression	What it means
Login validation	username_correct AND password_correct	User only logs in if both username AND password are correct
Alarm system	door_open OR window_open	Alarm triggers if door OR window (or both) are opened
NOT on standby	NOT standby_mode	Device is active when it is NOT in standby mode
Search filter	(price_low OR sale) AND in_stock	Show items that are (cheap OR on sale) AND available
Game unlock	level_complete AND NOT enemies_remaining	Level unlocks when complete AND no enemies left

Understanding Boolean logic helps you write better conditions in programs and understand how digital systems make decisions.

How to draw a logic circuit

Step 1: Identify the inputs (A, B, C, etc.) and draw them as lines on the left.
Step 2: Work from the **innermost brackets outward**, drawing gates for each operation.
Step 3: Connect the output of one gate to the input of the next.
Step 4: Label the final output (usually P or Q).
Step 5: Check: NOT gates have 1 input, AND/OR gates have 2 inputs.

1. Logic gates — the building blocks

A **logic gate** is an electronic component that takes one or more binary inputs and produces a single binary output based on a specific rule. The three gates you need to know for OCR GCSE are AND, OR, and NOT. These gates are the fundamental building blocks of all digital circuits, from simple calculators to complex processors.

The AND gate

The AND gate outputs 1 **only when all inputs are 1**. Think of it as a strict requirement — everything must be true for the output to be true. In programming, this is equivalent to using `AND` or `&&` to combine conditions.

Exam phrasing: "The AND gate gives an output of 1 when input A **and** input B are both 1."

Memory trick: AND is demanding — it needs **all** inputs to be 1. The only row with output 1 is the last row (1 AND 1 = 1).

The OR gate

The OR gate outputs 1 **when at least one input is 1**. It's more relaxed than AND — only one condition needs to be true. In programming, this is equivalent to using `OR` or `||` to combine conditions.

Exam phrasing: "The OR gate gives an output of 1 when input A **or** input B (or both) is 1."

Memory trick: OR is generous — it only needs **one** input to be 1. The only row with output 0 is the first row (0 OR 0 = 0).

The NOT gate

The NOT gate (also called an inverter) has only **one input** and outputs the **opposite** value. If the input is 1, the output is 0. If the input is 0, the output is 1. In programming, this is equivalent to using `NOT` or `!` before a condition.

Exam phrasing: "The NOT gate inverts the input — it gives an output of 1 when the input is 0, and an output of 0 when the input is 1."

Key symbol feature: The NOT gate has a small **circle (bubble)** at its output. This bubble indicates inversion. If you forget the bubble, you've drawn the wrong gate.

> **Exam tip — gate recognition.**
> When identifying gates in a circuit diagram:
> - **NOT:** Triangle with a bubble at the tip (1 input)
> - **AND:** D-shape — flat on the left, curved on the right (2 inputs)
> - **OR:** Shield/arrow shape — curved on both sides, pointed on the right (2 inputs)
>
> Count the inputs: if there's only one input line, it must be NOT.

Comparing AND and OR – key differences

Feature	AND gate	OR gate
Output is 1 when...	ALL inputs are 1	AT LEAST ONE input is 1
Output is 0 when...	ANY input is 0	ALL inputs are 0
Number of 1s in 2-input truth table	1 (only last row)	3 (all except first row)
Number of 0s in 2-input truth table	3	1 (only first row)
Programming equivalent	`if A AND B then`	`if A OR B then`
English meaning	"both must be true"	"either can be true"
Strictness	Strict (demanding)	Relaxed (generous)

> **Connection to programming.** Boolean logic is directly used in programming conditions. When you write `if age >= 18 AND hasTicket == true then`, both conditions must be true to enter. When you write `if isStudent == true OR isOAP == true then`, the discount applies if either condition is true. Understanding logic gates helps you write correct conditions in your programs.

Worked Example 1 — Basic NOT gate truth table

Question: Complete the truth table for the expression P = NOT A. [2 marks]

Step 1: NOT has one input, so there are $2^1 = 2$ rows.
Step 2: For each input value, output the opposite.

A	P = NOT A
0	1
1	0

Answer: When A = 0, P = 1. When A = 1, P = 0.

Exam tip: NOT always gives the opposite. Students sometimes write the same value as the input — always invert!

Worked Example 2 — Basic AND gate truth table

Question: Complete the truth table for the expression P = A AND B. [2 marks]

Step 1: AND has two inputs, so there are $2^2 = 4$ rows.
Step 2: Output 1 only when BOTH inputs are 1.

A	B	P = A AND B
0	0	0
0	1	0
1	0	0
1	1	1

Answer: Only the last row (1 AND 1) gives output 1.

Exam tip: For AND, remember "both must be 1". The output column has three 0s and one 1.

Worked Example 3 — Basic OR gate truth table

Question: Complete the truth table for the expression P = A OR B. [2 marks]

Step 1: OR has two inputs, so there are $2^2 = 4$ rows.
Step 2: Output 1 when AT LEAST ONE input is 1.

A	B	P = A OR B
0	0	**0**
0	1	**1**
1	0	**1**
1	1	**1**

Answer: Only the first row (0 OR 0) gives output 0.

Exam tip: For OR, remember "at least one must be 1". The output column has one 0 and three 1s — the opposite pattern to AND.

Worked Example 4 – Evaluating a simple expression

Question: What is the output of NOT (1 AND 0)? [1 mark]

Step 1: Evaluate the brackets first: 1 AND 0 = 0 (both must be 1 for AND).
Step 2: Apply NOT to the result: NOT 0 = 1.
Answer: 1

Exam tip: Always work from the innermost brackets outward, just like in maths.

2. Truth tables – combining gates

When Boolean expressions combine multiple gates, you need a systematic approach to complete truth tables. The key is to break down the expression into sub-expressions and evaluate them step by step. For expressions with 3 inputs (A, B, C), you'll have $2^3 = 8$ rows in your truth table.

Building truth tables systematically

For complex expressions, add **intermediate columns** for sub-expressions. This shows your working and helps you avoid errors. For example, if the expression is (A AND B) OR C:

- Column 1: A (input)
- Column 2: B (input)
- Column 3: C (input)
- Column 4: A AND B (intermediate)
- Column 5: (A AND B) OR C (final output)

The intermediate column isn't always required for marks, but it helps you get the right answer and may earn method marks if your final answer is wrong.

Listing input combinations

Always list input combinations in **binary counting order** to ensure you don't miss any. For 3 inputs:

A	B	C
0	0	0
0	0	1
0	1	0
0	1	1
1	0	0
1	0	1
1	1	0
1	1	1

Notice how C alternates every row, B alternates every 2 rows, and A alternates every 4 rows.

Expressions with NOT

When NOT appears in an expression, pay careful attention to **what it applies to**:

- `NOT A AND B` means `(NOT A) AND B` — NOT applies only to A
- `NOT (A AND B)` means NOT applies to the whole result of A AND B

These give **different results**. The brackets make a crucial difference. When NOT is outside brackets, evaluate the brackets first, then invert the result.

Exam tip — 3-input tables.
For 3 inputs, you need exactly **8 rows**. If your table has more or fewer rows, something is wrong. Common mistakes:

- Missing the 000 row (all zeros)
- Duplicating a row
- Only doing 4 rows (that's for 2 inputs)

Worked Example 5 — (A AND B) OR C [2024 exam style]

Question: Complete the truth table for P = (A AND B) OR C. [4 marks]

Step 1: Set up the table with 8 rows (3 inputs → 2^3 = 8 combinations).
Step 2: Add an intermediate column for (A AND B).
Step 3: For each row: first calculate A AND B, then OR that result with C.

A	B	C	A AND B	P = (A AND B) OR C
0	0	0	0	**0**
0	0	1	0	**1**
0	1	0	0	**0**

A	B	C	A AND B	P = (A AND B) OR C
0	1	1	0	**1**
1	0	0	0	**0**
1	0	1	0	**1**
1	1	0	1	**1**
1	1	1	1	**1**

Check: Row 7 — A=1, B=1, so A AND B = 1. Then 1 OR 0 = 1. ✓

Exam tip: This exact question appeared in the 2024 OCR exam. Notice that the output is 1 whenever C=1 OR when both A and B are 1.

Worked Example 6 – NOT A AND B

Question: Complete the truth table for P = NOT A AND B. [2 marks]

Step 1: Identify precedence: NOT applies to A only (NOT A AND B = (NOT A) AND B).
Step 2: First invert A, then AND with B.

A	B	NOT A	P = NOT A AND B
0	0	1	**0**
0	1	1	**1**
1	0	0	**0**
1	1	0	**0**

Answer: P = 1 only when A = 0 AND B = 1.

Exam tip: Without brackets, NOT has highest precedence. It applies only to the variable immediately after it.

Worked Example 7 – NOT (A OR B)

Question: Complete the truth table for P = NOT (A OR B). [2 marks]

Step 1: The brackets mean: first calculate A OR B, then invert the result.
Step 2: Add intermediate column for A OR B.

A	B	A OR B	P = NOT (A OR B)
0	0	0	**1**
0	1	1	**0**
1	0	1	**0**
1	1	1	**0**

Answer: P = 1 only when BOTH A and B are 0.

Exam tip: Compare with Example 6. The brackets completely change the meaning. NOT (A OR B) ≠ NOT A OR B.

Worked Example 8 — A OR (B AND NOT C)

Question: Complete the truth table for P = A OR (B AND NOT C). [4 marks]

Step 1: Work out precedence: NOT C first, then B AND (NOT C), then OR with A.
Step 2: Add intermediate columns.

A	B	C	NOT C	B AND NOT C	P
0	0	0	1	0	**0**
0	0	1	0	0	**0**
0	1	0	1	1	**1**
0	1	1	0	0	**0**
1	0	0	1	0	**1**
1	0	1	0	0	**1**
1	1	0	1	1	**1**
1	1	1	0	0	**1**

Check: Row 3 — NOT 0 = 1, 1 AND 1 = 1, 0 OR 1 = 1. ✓

Exam tip: When you have 3 operations, use intermediate columns. It makes marking easier and you're less likely to make errors.

3. Logic circuits — drawing and interpreting diagrams

Logic circuits show how gates are connected together to implement Boolean expressions. In the exam, you may be asked to **draw a circuit from an expression** or to **work out the expression from a given circuit**. The key skill is translating between the written expression and the visual diagram.

Drawing circuits from expressions

To draw a circuit for an expression like `NOT A AND (B OR C)`:

1. **Identify the main operation** — here it's AND (combining NOT A with the result of B OR C)
2. **Draw inputs on the left** — lines labelled A, B, C
3. **Add gates for sub-expressions** — NOT gate for A, OR gate for B and C
4. **Connect to the main gate** — AND gate taking two inputs
5. **Label the output** — P on the right

Work from the **innermost brackets outward**. The last gate you draw should be the **outermost operation**.

Reading expressions from circuits

To work out the expression from a circuit:

1. **Start at the output** — identify the final gate
2. **Trace back** — what feeds into this gate?
3. **Build the expression** — the final gate becomes the outermost operation
4. **Add brackets** — for any sub-circuits that feed into a gate

If you see a **bubble** (small circle) on a wire, that indicates NOT is applied at that point.

Common circuit patterns

Pattern 1: NOT before AND

For `NOT A AND B`:

- A → NOT gate → into AND gate (top input)
- B → directly into AND gate (bottom input)
- AND gate → P

Pattern 2: Operation then NOT

For `NOT (A AND B)`:

- A and B → into AND gate
- AND gate → NOT gate → P

> **Exam tip — circuit drawing checklist.**
> Before finishing your circuit, check:
> - ✓ All inputs are labelled (A, B, C)
> - ✓ Output is labelled (P)
> - ✓ NOT gates have exactly 1 input
> - ✓ AND/OR gates have exactly 2 inputs
> - ✓ NOT gates have the bubble at the output
> - ✓ Gates are the correct shape
> - ✓ Wires connect clearly (no ambiguous crossings)

Common exam expressions and their structures

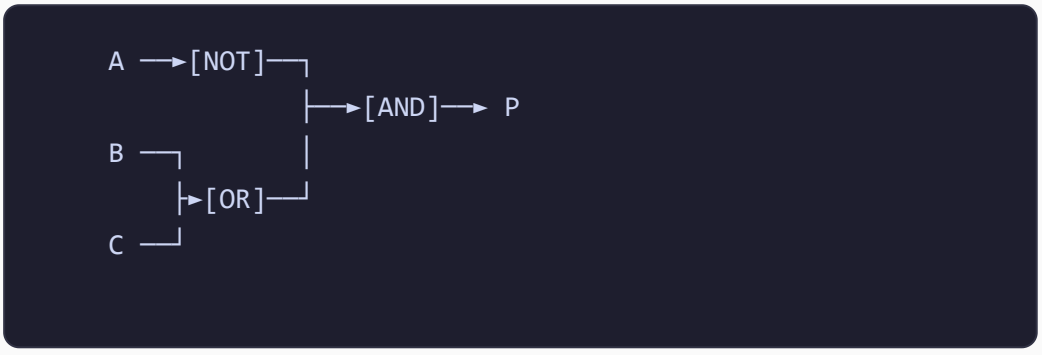
Expression	Number of gates needed	Gate types	Structure hint
NOT A AND B	2	1 NOT, 1 AND	NOT on A, then AND with B
A OR NOT B	2	1 NOT, 1 OR	NOT on B, then OR with A
NOT (A AND B)	2	1 AND, 1 NOT	AND first, then NOT the result
(A AND B) OR C	2	1 AND, 1 OR	AND on A,B; OR result with C
NOT A AND (B OR C)	3	1 NOT, 1 OR, 1 AND	NOT on A; OR on B,C; AND the two results
(A OR B) AND (NOT C)	3	1 OR, 1 NOT, 1 AND	OR on A,B; NOT on C; AND the two results

Exam insight — 2024 paper analysis. The 2024 OCR J277/02 exam asked students to complete a truth table for (A AND B) OR C [4 marks] and draw a logic circuit for NOT A AND (B OR C) [3 marks]. Both questions are typical of the difficulty level. Practice completing 8-row truth tables quickly and drawing 3-gate circuits accurately.

Worked Example 9 — Draw circuit for NOT A AND (B OR C) [2024 exam]

Question: Draw a logic circuit for P = NOT A AND (B OR C). [3 marks]

Step 1: Identify the structure: AND combines (NOT A) with (B OR C).
Step 2: Draw three inputs on the left: A, B, C.
Step 3: A goes through a NOT gate.
Step 4: B and C go into an OR gate.
Step 5: The NOT output and OR output go into an AND gate.
Step 6: Label the AND output as P.



Key points: NOT gate has the bubble. OR gate is curved/shield shape. AND gate is D-shape.

Exam tip: This exact question was in the 2024 OCR exam. Many students lost marks for missing the NOT bubble or using wrong gate shapes.

Worked Example 10 — Determine output for given inputs

Question: For the circuit P = NOT A AND (B OR C), determine the output when A = 1, B = 0, C = 1. [2 marks]

Step 1: Calculate NOT A: NOT 1 = 0
Step 2: Calculate B OR C: 0 OR 1 = 1
Step 3: Calculate the AND: 0 AND 1 = 0
Answer: P = 0
Verification: For the output to be 1, we need NOT A = 1 (so A must be 0) AND (B OR C) = 1 (so at least one of B, C must be 1). Since A = 1, NOT A = 0, so the AND gate outputs 0 regardless of the other input.

Exam tip: Work through the circuit systematically. Write down each intermediate result to avoid errors.

Worked Example 11 — Identify expression from circuit

Question: A logic circuit has inputs A and B. A goes into a NOT gate. The NOT output and B go into an OR gate. Write the Boolean expression for the output P. [2 marks]

Step 1: Trace the circuit: A → NOT → result is NOT A
Step 2: NOT A and B go into OR gate
Step 3: Write the expression: P = NOT A OR B
Or equivalently: P = (NOT A) OR B
Answer: P = NOT A OR B

Exam tip: When reading circuits, start at the output and work backwards. The final gate tells you the outermost operation.

Worked Example 12 — Complex 3-input circuit analysis

Question: For the expression P = (A OR B) AND (NOT C), complete the truth table and identify all input combinations that give output 1. [5 marks]

Step 1: Set up truth table with intermediate columns.

A	B	C	A OR B	NOT C	P
0	0	0	0	1	**0**
0	0	1	0	0	**0**
0	1	0	1	1	**1**
0	1	1	1	0	**0**
1	0	0	1	1	**1**
1	0	1	1	0	**0**
1	1	0	1	1	**1**
1	1	1	1	0	**0**

Step 2: Identify rows where P = 1: rows 3, 5, and 7.
Answer: Output is 1 when:

- A = 0, B = 1, C = 0
- A = 1, B = 0, C = 0
- A = 1, B = 1, C = 0

Pattern: P = 1 when (A OR B is true) AND (C is false). In other words: at least one of A or B must be 1, AND C must be 0.

Exam tip: Higher-mark questions often ask you to identify patterns. Look for what the input combinations that give output 1 have in common.

Summary of worked examples

#	Topic	Expression/Task	Key skill
1	NOT gate	P = NOT A	Basic truth table, inversion
2	AND gate	P = A AND B	Basic truth table, "both must be 1"
3	OR gate	P = A OR B	Basic truth table, "at least one"
4	Simple evaluation	NOT (1 AND 0)	Brackets first, then NOT
5	3-input truth table	P = (A AND B) OR C	2024 exam style, 8 rows
6	NOT with AND	P = NOT A AND B	Precedence without brackets

#	Topic	Expression/Task	Key skill
7	NOT with brackets	P = NOT (A OR B)	NOT applies to whole expression
8	Complex expression	P = A OR (B AND NOT C)	Multiple intermediate steps
9	Circuit drawing	P = NOT A AND (B OR C)	2024 exam, gate shapes
10	Circuit evaluation	Determine output for inputs	Trace through circuit
11	Circuit reading	Write expression from circuit	Trace from output backwards
12	Pattern analysis	P = (A OR B) AND (NOT C)	Identify when output is 1

Quick reference

Key terms

Term	Definition (exam-answer style)
Boolean logic	A form of algebra where all values are either TRUE (1) or FALSE (0)
Logic gate	An electronic component that takes binary inputs and produces a binary output based on a specific rule
AND gate	Outputs 1 only when all inputs are 1; otherwise outputs 0
OR gate	Outputs 1 when at least one input is 1; outputs 0 only when all inputs are 0
NOT gate	Inverts the input; outputs 1 when input is 0, outputs 0 when input is 1
Truth table	A table showing all possible input combinations and their corresponding outputs
Logic circuit	A diagram showing how logic gates are connected to implement a Boolean expression
Boolean expression	A formula using AND, OR, NOT to describe a logical operation
Precedence	The order in which operations are evaluated: NOT first, then AND, then OR
Inverter	Another name for a NOT gate; inverts (reverses) the input signal

Gate quick reference

Gate	Inputs	Symbol description	Rule	Memory trick
AND	2	D-shape (flat left, curved right)	Both inputs must be 1	Only one 1 in output column (last row)
OR	2	Curved shield (pointed right)	At least one input must be 1	Only one 0 in output column (first row)
NOT	1	Triangle with bubble at output	Output is opposite of input	The bubble means "invert"

Truth table row counts

Number of inputs	1	2	3	4
Number of rows	2	4	8	16
Formula	2^1	2^2	2^3	2^4

Precedence order

Priority	Operation	Evaluated
1 (highest)	Brackets	First
2	NOT	Second
3	AND	Third
4 (lowest)	OR	Last

Typical pitfalls

Pitfall 1 — Missing the NOT bubble.
Wrong: Drawing NOT as a plain triangle.
Right: NOT gate must have a **small circle (bubble)** at its output. Without the bubble, you've drawn a buffer, not a NOT gate.

Pitfall 2 — Wrong number of inputs.
Wrong: Drawing AND or OR with 3 inputs, or NOT with 2 inputs.
Right: NOT has **1 input**. AND and OR have **2 inputs** each (in OCR GCSE). To combine 3 values, you need multiple gates.

Pitfall 3 — Confusing AND and OR patterns.
Wrong: Thinking AND gives mostly 1s.
Right: AND is strict — **only one 1** in output (when all inputs are 1). OR is generous — **only one 0** in output (when all inputs are 0).

Pitfall 4 — Wrong precedence for NOT.
Wrong: Treating NOT A AND B as NOT (A AND B).
Right: Without brackets, NOT applies **only to the next variable**. NOT A AND B = (NOT A) AND B.

Pitfall 5 — Wrong row count.
Wrong: Only 4 rows for a 3-input truth table.
Right: For n inputs, you need **2^n rows**. 3 inputs = 8 rows. Always use binary counting order.

Pitfall 6 — Confusing gate symbols.
Wrong: Drawing AND with a curved left side.
Right: AND is D-shaped (**flat left, curved right**). OR is shield-shaped (**curved both sides, pointed right**).

Pitfall 7 — NOT outputs the same as input.
Wrong: NOT 1 = 1.
Right: NOT **always inverts**. NOT 1 = 0. NOT 0 = 1. If your output matches your input, you've made an error.

Pitfall 8 — Missing intermediate columns.
Wrong: Trying to calculate (A AND B) OR C in one step.
Right: Add an **intermediate column** for A AND B, then use that result with C. Shows working and reduces errors.

Pitfall 9 — Unlabelled circuits.
Wrong: Drawing gates but not labelling inputs A, B, C or output P.
Right: Always label inputs (on the left) and output (on the right). Examiners need to know which input is which.

Pitfall 10 — Thinking OR means "exclusive or".
Wrong: 1 OR 1 = 0 (thinking only one can be true).
Right: OR in Boolean logic is **inclusive**. 1 OR 1 = 1. "At least one" includes "both".

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